Thejaswini T

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	8.22
VELAMMAL VIDYALAYA MUTHUTHEVANPATTI THENI	Class XII (CBSE)	2023	82.80%
VELAMMAL VIDYALAYA MUTHUTHEVANPATTI THENI	Class X (CBSE)	2021	94.40%

SCHOLASTIC ACHIEVEMENTS

certificate holder in NSO(national science olympiad),IMO(international mathematics olympiad)

took part in olympiads such as IMO and NSO and won certificates

International General Knowledge Olympiad |IGKO Bronze medal holder :

Won a medal in the International General Knowledge Olympiad (IGKO), demonstrating strong general knowledge

POSITIONS OF RESPONSIBILITY

Cultural Council - Branding & Marketing Coordinator:

- Managed branding, promotional activities, and social media outreach for cultural events and initiatives.

Alumni Ideation & Operations Coordinator:

Coordinated alumni initiatives from ideation to execution, supporting planning, operations, and event coordination.

INTERNSHIP EXPERIENCE

AI Research Intern | DeepDrishti AI

- Contributed to AI-driven projects across **industrial maintenance, catalyst intelligence, and pharmaceutical knowledge management**.
- Worked on **Catalyst Intelligence AI**, supporting catalyst data organization, performance analysis, and recommendation workflows using parameters such as conversion, selectivity, temperature, and pressure.
- Contributed to **PharmaKnowHub** by validating pharmaceutical relationships, including product-manufacturer, raw material-product, and CAS number linkages.
- Designed and implemented the **workflow component** of a maintenance automation project and created the associated dataset and data-storage structure.
- Applied **Python, machine learning, data analysis, prompt engineering, and AI-assisted tools** across project workflows.
- Gained exposure to **pharmaceutical regulatory research** and scientific information analysis during the internship.

PROJECTS

Undergraduate research project - Solid state pharmaceutical chemistry|Flextal labs|Prof.C Malla Reddy

IIT Hyderabad,Dept of Chemistry

- Designed and executed polymorph screening experiments using **various Active Pharmaceutical Ingredients**

(APIs) and a range of **antisolvents** to identify novel solid forms.

- Characterized polymorphs using **Powder X-Ray Diffraction (PXRD)**
- Documented and analyzed **crystallization outcomes** across different APIs to correlate molecular structure with polymorphic behavior.
- Utilized **antisolvent precipitation techniques** ,cooling crystallization techniques and slurry method to produce distinct crystal habits
- Gained proficiency in **solvent evaporation**, **slow cooling**, and **grinding** methods
- Designed and synthesized co-crystals using supramolecular synthesis technique

Catalyst Intelligence AI|*Python, AI/ML, Prompt Engineering, Git***|**

- Contributed to an AI-based catalyst recommendation and ranking system using structured scientific data.

- Worked with prompt-driven workflows to process catalyst properties and user-defined objectives.

- Analyzed and validated model outputs using conversion, selectivity, yield, temperature and pressure parameters.

- Iterated on outputs and data processing to improve the relevance of catalyst recommendations.

PharmaKnowHub|*Python, MongoDB, AI/ML, Prompt Engineering***|**

- Contributed to an AI-powered pharmaceutical knowledge system for mapping relationships across products, manufacturers, raw materials and CAS numbers.

- Used prompt-based workflows to support information extraction and relationship identification.

- Validated AI-generated relationships against structured pharmaceutical data and identified incorrect or missing linkages.

- Supported iterative improvement of knowledge coverage and relationship validation.

Maintenance Automation|*Python, n8n, MongoDB, Docke*r**|**

- Enhanced skills in deep learning, RNNs, and sequential data modeling

- Created workflow components for an AI-assisted maintenance automation system.

- Built and organized datasets to support automated processing of maintenance-related information.

- Used **n8n** to connect workflow stages and automate data-processing tasks.

- Worked with **MongoDB** for data storage and **Docker** for the application environment.

RELEVANT COURSES

chemistry :

Physical Chemistry: Thermodynamics, Chemical Kinetics, Electrochemistry

Organic Chemistry: Organic Reaction Mechanisms, Chemical Synthesis, Stereochemistry, Drug Design

Inorganic Chemistry: Coordination Chemistry

Industrial & Materials Chemistry: Polymer Science, Materials Science, Separation Processes

Pharmaceutical & Computational Chemistry: Biochemistry, ML in Bioinformatics,

SKILLS

Research: Scientific Literature Analysis, Experimental Design, Technical Documentation

- **laboratory skills :** Wet chemistry techniques, titration, crystallization ,Sample preparation ,Basic synthesis and compound purification,Solid-state and co-crystal preparation methods

EXTRACURRICULAR

- Trained Dancer : Performed in multiple cultural programs; experienced in classical and contemporary styles and have won many prizes
- Keyboard Player : performed at and local music events and won prizes in events
- Badminton and table tennis player : enjoy palying both badminton and table tennis
- Olympiad Participant - participated in many olympiads such as IMO,IGKO,NSO and YLE